Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/14/2024
PI Name:	Walsh, Kenneth Ph.D.		
Project Title:	Space Radiation Exposure and Risk Mediated by Clonal Hematopoiesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kw9ar@virginia.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	434-243-8303
Organization Name:	University of Virginia, Charlottesville		
PI Address 1:	Hematovascular Biology Center		
PI Address 2:	415 Lane Rd, PO Box 801394, Suite 1010		
PI Web Page:			
City:	Charlottesville	**State:**	VA
Zip Code:	22903-3390	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	01/29/2021	**End Date:**	01/28/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI): Eunbee Park has left the project. We have 2 bachelor of science graduates and an MD/PhD student working in part on this project.		
COI Name (Institution):	Garrett-Bakelman, Francine M.D., Ph.D. (University of Virginia, Charlottesville) Hirschi, Karen Ph.D. (Yale University) Goukassian, David M.D., Ph.D. (ICAHN School of Medicine at Mount Sinai) Evans, Megan A (University of Virginia)		
Grant/Contract No.:	80NSSC21K0549		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During spaceflight, astronauts are exposed to many stresses that alter multiple physiological systems. The recent NASA Twins Study provided a highly detailed analysis of how prolonged, low orbit space travel may contribute to genotoxic stress, elevated DNA damage responses and genomic instability in leukocytes. The observed genomic instability during and after flight suggests that the ionizing radiation exposure caused DNA damage to hematopoietic stem cells that replenish blood cells throughout life. Thus, it is conceivable that these alterations will contribute to the development of hematologic malignancies and other chronic diseases through changes in immune cell function. Furthermore, these effects may be particularly magnified when traveling beyond Earth's geomagnetic field where there is increased exposure to high atomic number and high energy radiation. Recent epidemiological studies have documented the prevalence of somatic mutations within the cells of the hematopoietic system in healthy individuals. These acquired DNA mutations accumulate with age and, in some instances, can provide a competitive advantage to the mutant cell thus allowing for its clonal expansion. This phenomenon is known as clonal hematopoiesis of indeterminate potential (CHIP). While the mutational landscape of CHIP has only partially been deciphered, some of these clonal expansions can be attributed to somatic mutations in driver genes that are recurrently mutated in blood malignancies. These driver genes include epigenetic regulators (TET2, DNMT3A, ASXL1), spliceosome components (SF3B1, SRSF2), signaling proteins (JAK2), and DNA damage response molecules (TP53, PPM1D). Studies show that CHIP is associated with an increased risk of all-cause mortality. While there is a marked increase in the frequency of hematological cancer in individuals with CHIP, which is to be expected, the major cause of the increased mortality in these populations appears to be an increase in cardiovascular diseases including coronary heart disease, stroke, and early-onset myocardial infarction. Studies in the applicants' laboratories have provided evidence for a causal link between CHIP, derived from mutations in TET2, DNMT3A, JAK2, TP53 and PPM1D genes, and cardiovascular, metabolic, and renal pathologies. In some instances, it was shown that the pathological effects of a CHIP driver mutation (TET2, TP53 and PPM1D) could be mitigated with specific anti-inflammatory drugs. Of particular relevance to the proposed studies, there is an accelerated form of clonal hematopoiesis that is observed in individuals that have undergone myelosuppressive treatment and is referred to as "therapy-related clonal hematopoiesis." Under these conditions, it has been shown that there are hematopoietic clonal expansions with a very high frequency of mutations in PPM1D andTP53, both of which are classic DNA damage response genes. In individuals undergoing cytotoxic therapy, the hematopoietic system is likely under extreme stress, and it is thought that mutations in genes such as TP53 and PPM1D confer the mutated hematopoietic stem cell with a survival advantage against genotoxic stress induced by chemotherapy. Recent work from the applicants' laboratories have shown that this form of CHIP can synergize with the genotoxic agent's direct effect on the cardiovascular system to promote a more robust cardiomyopathic phenotype. While the impact of space travel on CHIP is completely unknown, it is reasonable to speculate that space radiation in combination with other space travel-related stresses will lead to radiation-specific and gene-specific accelerations of clonal hematopoiesis. Further, these forms of CHIP may increase the risk of leukemogenic and cardiovascular pathologies in a radiation- and gene-specific manner.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While the impact of space travel on CHIP is completely unknown, it is reasonable to speculate that space radiation in combination with other space travel-related stresses will lead to radiation-specific and gene-specific accelerations of clonal hematopoiesis. Further, these forms of CHIP may increase the risk of leukemogenic and cardiovascular pathologies in a radiation- and gene-specific manner.
Task Progress:	The accumulation of somatic mutations is a phenomenon that occurs in all tissues throughout the body during aging. In the hematopoietic system, a somatic mutation can randomly occur in a preleukemic driver gene within a hematopoietic stem and progenitor cell, providing that cell with a competitive advantage and thereby facilitating a clonal expansion. This condition is known as clonal hematopoiesis (CH) and leads to a population of mutant leukocytes in the peripheral blood. Studies suggest that these mutations may alter the immunological profile of leukocytes and can thereby causally contribute to age-related diseases. Thus, understanding how ionizing space radiation supports clone growth and the level at which clones become pathological is of paramount importance. During the current year we were able to participate in the Fall 2024 NASA Space Radiation Laboratory (NSRL) campaign. In the months leading up to this campaign, we prepared the mice that were used as bone marrow donors. We purchased 96 recipient mice from Jackson Laboratory. These recipients received either DNMT3A KO bone marrow cells (Dnmt3a fl/fl, vav-cre+) or DNMT3A wild-type bone marrow cells (Dnmt3a fl/fl, vav-cre-) via an adoptive transfer bone marrow transplant (BMT) approach. Approximately 2 months after bone marrow transplantation, the mice were transported to Brookhaven National Laboratory (BNL) and exposed to one of four types of radiation: no radiation, 100cGy gamma, 100cGy simGCRsim, or 100cGy SPEsim. One member of the Walsh lab traveled to BNL to complete these irradiation sessions. After the irradiation sessions, the mice were transported back to the University of Virginia and we will perform serial blood sampling and echocardiography for the next 18 months. Currently this DNMT3A cohort is a few weeks post radiation exposure. Thus far, this cohort of mice has had baseline sampling for flow cytometry and whole blood analysis. Echocardiography was performed before irradiation. Body weights will be measured monthly. In addition, the TET2 cohort completed the study time course and tissues were collected in December 2023/January 2024. We have also continued to monitor and perform serial sampling for the TP53 wild-type cohort which is currently 17 months post radiation exposure. Furthermore, we have performed some additional analyses for the original TP53 mutant male cohort, including histology, which was analyzed by a veterinary pathologist – and have evaluated the mosaic loss of the Y chromosome in splenic tissue. Going forward, we plan to continue flow cytometry and whole blood analysis every 4-6 months and echocardiography every 6-9 months for the DNMT3A cohort. The mice remaining in the repeat TP53 wild-type cohort will be euthanized and tissues harvested in December 2024. We are also in the early stages of drafting a manuscript that describes our findings on the models of TP53, PPM1D, and TET2 clonal hematopoiesis. TP53 and PPM1D are "therapy-related" clonal hematopoiesis (t-CH) driver genes, and they respond to genotoxic stress (such as simulated space radiation). TET2 is a classical "age-related" clonal hematopoiesis driver gene. The analysis of the TET2 model provides a contrast to the t-CH driver genes.

Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Walsh K, Cochran JD, Evans MA. "Clonal hematopoiesis: Getting to the heart of the problem with clone size." JACC Heart Fail. 2024 May;12(5):915-17. https://doi.org/10.1016/j.jchf.2023.12.009 ; PubMed PMID: 38300211; PubMed Central PMCID: PMC11081812 , May-2024
Articles in Peer-reviewed Journals	Sano S, Walsh K. "Can dental problems be indicative of clonal hematopoiesis?" Cell Res. 2024 Jul 25. Epub ahead of print. https://doi.org/10.1038/s41422-024-01006-2 ; PubMed PMID: 39054344. , Jul-2024
Articles in Peer-reviewed Journals	Thel MC, Cochran JD, Teruya S, Xie CR, Srinivasan R, Chavkin NW, Arai Y, Sano S, Mirabal Santos A, De Los Santos J, Fine D, Sabogal N, Ullah I, Helmke S, Rodriguez C, Prokaeva T, Foster RH, Spencer BH, Izumiya Y, Maurer MS, Walsh K, Ruberg FL. "Mosaic loss of the Y chromosome is enriched in patients with wild-type transthyretin (ATTRwt) cardiac amyloidosis and associated with increased mortality." Circ Heart Fail. 2024 Aug;17(8):e011681. Review. https://doi.org/10.1161/CIRCHEARTFAILURE.124.011681 ; PubMed PMID: 38853760 ; PubMed Central PMCID: PMC11335443 , Aug-2024
Articles in Peer-reviewed Journals	Polizio AH, Marino L, Min KD, Yura Y, Rolauer L, Cochran JD, Evans MA, Park E, Doviak H, Miura-Yura E, Good ME, Wolpe AG, Grandoch M, Isakson B, Walsh K. "Experimental TET2 clonal hematopoiesis predisposes to renal hypertension through an inflammasome-mediated mechanism." Circ. Res. 2024 Oct 11;135(9):933-50. https://doi.org/10.1161/CIRCRESAHA.124.324492 ; PubMed PMID: 39234670; PubMed Central PMCID: PMC11519839 , Oct-2024
Articles in Peer-reviewed Journals	Verdonschot JAJ, Fuster JJ, Walsh K, Heymans SRB. "The emerging role of clonal haematopoiesis in the pathogenesis of dilated cardiomyopathy." Eur Heart J. 2024 Dec 1;45(45):4797-807. Review. https://doi.org/10.1093/eurheartj/ehae682 ; PubMed PMID: 39417710; PubMed PMCID: PMC11638724 , Dec-2024
Articles in Peer-reviewed Journals	Cochran JD, Walsh K. "Clonal hematopoiesis: The emergent CVD risk factor." Arterioscler Thromb Vasc Biol. 2024 Mar 27;44(4):768-71. https://doi.org/10.1161/ATVBAHA.123.319562 ; PMID: 38536898; PMCID: PMC10977652 , Mar-2024
Articles in Peer-reviewed Journals	Horitani K, Chavkin NW, Arai Y, Wang Y, Ogawa H, Yura Y, Evans MA, Cochran JD, Thel MC, Polizio AH, Sano M, Miura-Yura E, Arai Y, Doviak H, Arnold AP, Gelfand BD, Hirschi KK, Sano S, Walsh K. "Disruption of the Uty epigenetic regulator locus in hematopoietic cells phenocopies the profibrotic attributes of Y chromosome loss in heart failure." Nat Cardiovasc Res. 2024 Mar 8. https://doi.org/10.1038/s44161-024-00441-z ; PMID: 39183958; PMCID: PMC11343478 , Mar-2024